Number Theory

Divisibility, the Division Algorithm, Modular Arithmetic, Prime Numbers, and the Euclidean Algorithm

1 Divisibility

Number theory is the branch of mathematics devoted to the study of the integers $\mathbb{Z} = \{..., -2, -1, 0, 1, 2, ...\}$ and their arithmetic properties. While historically regarded as a purely theoretical discipline, number theory forms the backbone of modern computer science—powering public-key cryptography (such as RSA and Diffie-Hellman), cryptographic hash functions, pseudorandom number generation, error-correcting codes, and hash table indexing.

The most basic relationship between two integers is whether one divides the other evenly without leaving a remainder.

DEFINITION **Divisibility**

Let a and b be integers with $a \neq 0$. We say that a **divides** b if there exists an integer $c \in \mathbb{Z}$ such that:

$$b = ac$$

- When a divides b, we call a a **factor** or **divisor** of b, and we call b a **multiple** of a.
- Notation:
  - $a \mid b$ denotes that a divides b.
  - $a \nmid b$ denotes that a does not divide b.

INTUITION & DEEP DIVE **Why Divisibility Uses Multiplication Instead of Fractions**

Notice that the definition $b = ac$ (for some $c \in \mathbb{Z}$) is written entirely using integer multiplication rather than division ($\frac{b}{a}$). When working with integers, we want to stay inside $\mathbb{Z}$ without introducing fractions or floating-point numbers. Asking whether $a \mid b$ is simply asking whether the equation $ax = b$ has an integer solution $x \in \mathbb{Z}$.

EXAMPLE & WALKTHROUGH **Basic Examples of Divisibility**

1. **Does 5 divide 10?** Yes, $5 \mid 10$ because there is an integer $c = 2 \in \mathbb{Z}$ such that $10 = 5 \cdot 2$.
2. **Does 3 divide 6?** Yes, $3 \mid 6$ because there is an integer $c = 2 \in \mathbb{Z}$ such that $6 = 3 \cdot 2$.
3. **Does 2 divide 9?** No, $2 \nmid 9$ because no integer $c \in \mathbb{Z}$ satisfies $9 = 2c$ (since $2 \cdot 4 = 8 < 9 < 10 = 2 \cdot 5$).

1.1 Algebraic Properties of Divisibility

Divisibility obeys three foundational rules that let us combine, scale, and chain divisibility relationships without computing explicit quotients.

CORE CONCEPT **Theorem: Fundamental Properties of Divisibility**

Let a, b, and c be integers, where $a \neq 0$. Then:

1. **Additive Closure**: If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
2. **Scaling by Any Integer**: If $a \mid b$, then $a \mid bc$ for all integers $c \in \mathbb{Z}$.
3. **Transitivity**: If $a \mid b$ and $b \mid c$ (with $b \neq 0$), then $a \mid c$.

Proof:

- **(i)** Suppose $a \mid b$ and $a \mid c$. By definition, there exist integers $s, t \in \mathbb{Z}$ such that $b = as$ and $c = at$. Adding these two equations gives:

$$b + c = as + at = a(s + t)$$

  Since $s + t \in \mathbb{Z}$, it follows that $a \mid (b + c)$.
- **(ii)** Suppose $a \mid b$. Then there exists $k \in \mathbb{Z}$ such that $b = ak$. Multiplying both sides by any integer $c \in \mathbb{Z}$ gives:

$$bc = (ak)c = a(kc)$$

  Since $kc \in \mathbb{Z}$, we conclude that $a \mid bc$.
- **(iii)** Suppose $a \mid b$ and $b \mid c$. Then there exist integers $s, t \in \mathbb{Z}$ such that $b = as$ and $c = bt$. Substituting $b = as$ into the second equation gives:

$$c = (as)t = a(st)$$

  Since $st \in \mathbb{Z}$, we conclude that $a \mid c$. □

Combining parts (i) and (ii) immediately gives a useful **Linear Combination Property**: if $a \mid b$ and $a \mid c$, then $a \mid (mb + nc)$ for any integers $m, n \in \mathbb{Z}$.

EXAMPLE & WALKTHROUGH **Applying the Divisibility Properties**

- **Property (i)**: Since $2 \mid 6$ and $2 \mid 4$, we immediately know that $2 \mid (6 + 4)$, or $2 \mid 10$.
- **Property (ii)**: Since $5 \mid 15$, multiplying 15 by $c = 4$ guarantees that $5 \mid (15 \cdot 4)$, or $5 \mid 60$.
- **Property (iii)**: Since $7 \mid 21$ and $21 \mid 42$, transitivity guarantees that $7 \mid 42$.

EXAMPLE & WALKTHROUGH **Practice Problems: Testing and Applying Divisibility**

Part A — Determine whether $a \mid b$ for the following pairs:

1. $a = 3, b = 10$: We check whether there is an integer $c \in \mathbb{Z}$ such that $10 = 3c$. Since $3 \cdot 3 = 9$ and $3 \cdot 4 = 12$, no such integer exists ($\frac{10}{3} = 3.33\ldots \notin \mathbb{Z}$). Therefore, $\mathbf{3 \nmid 10}$.
2. $a = 19, b = 95$: We check whether 95 is an integer multiple of 19. Since $95 = 19 \cdot 5$ and $5 \in \mathbb{Z}$, we conclude that $\mathbf{19 \mid 95}$.

Part B — Verify the following using the divisibility properties:

1. **If $8 \mid 32$ and $8 \mid 64$, does $8 \mid 96$? Yes.** By Property (i) (with $a = 8$, $b = 32$, and $c = 64$), since $8 \mid 32$ and $8 \mid 64$, we must have $8 \mid (32 + 64)$. Since $32 + 64 = 96$, it follows that $\mathbf{8 \mid 96}$ (indeed, $96 = 8 \cdot 12$).
2. **If $16 \mid 80$, does $16 \mid 240$? Yes.** Notice that $240 = 80 \cdot 3$. By Property (ii) (with $a = 16$, $b = 80$, and multiplier $c = 3 \in \mathbb{Z}$), since $16 \mid 80$, we have $16 \mid (80 \cdot 3)$, which means $\mathbf{16 \mid 240}$ (indeed, $240 = 16 \cdot 15$).
3. **If $8 \mid 24$ and $24 \mid 72$, does $8 \mid 72$? Yes.** By Property (iii) (transitivity, with $a = 8$, $b = 24$, and $c = 72$), since $8 \mid 24$ and $24 \mid 72$, it follows directly that $\mathbf{8 \mid 72}$ (indeed, $72 = 8 \cdot 9$).

PITFALL / WARNING **The Converse of Properties (i) and (ii) Does Not Hold**

Be careful not to reverse the implications in the divisibility theorem:

- **Sum trap**: $a \mid (b + c)$ does **not** imply that $a \mid b$ and $a \mid c$. For instance, $5 \mid (2 + 3)$, yet $5 \nmid 2$ and $5 \nmid 3$.
- **Product trap**: $a \mid bc$ does **not** generally imply $a \mid b$ or $a \mid c$ when a is composite. For example, $6 \mid (3 \cdot 4)$ because $6 \mid 12$, yet $6 \nmid 3$ and $6 \nmid 4$. (This implication only holds when a is a prime number, a result known as Euclid's Lemma.)

2 The Division Algorithm & Modulo Concept

When a positive integer $d > 0$ does not divide a evenly, integer division yields both an integer quotient and a strictly smaller non-negative remainder.

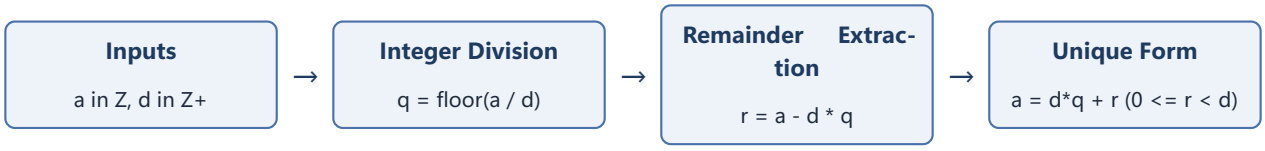
CORE CONCEPT **Theorem: The Division Algorithm**

Let a be an integer and d be a positive integer ($d \in \mathbb{Z}^+$). Then there exist **unique** integers q and r, with $0 \leq r < d$, such that:

$$a = dq + r$$

The four parts of this equation have standard names:

- a: **dividend**
- d: **divisor** (strictly positive: $d > 0$)
- q: **quotient**, denoted $q = a \div d = \left\lfloor \frac{a}{d} \right\rfloor$
- r: **remainder**, denoted $r = a \bmod d$, satisfying $0 \leq r < d$



Decomposition of an integer dividend a by positive divisor d via the Division Algorithm.

EXAMPLE & WALKTHROUGH **Division Algorithm Decompositions**

1. $10 = 4 \cdot 2 + 2$: Dividing $a = 10$ by $d = 4$ gives quotient $q = 2$ and remainder $r = 2$ (with $0 \leq 2 < 4$).
2. $90 = 7 \cdot 12 + 6$: Dividing $a = 90$ by $d = 7$ gives quotient $q = 12$ and remainder $r = 6$ (with $0 \leq 6 < 7$).
3. $25 = 6 \cdot 4 + 1$: Dividing $a = 25$ by $d = 6$ gives quotient $q = 4$ and remainder $r = 1$ (with $0 \leq 1 < 6$).

EXAMPLE & WALKTHROUGH **Practice Problems: Expressing Numbers as a = d * q + r**

Problem: Write each pair in the form $a = d \cdot q + r$ with $0 \leq r < d$:

1. $\boldsymbol{a = 91, d = 36}$: Dividing 91 by 36, the largest multiple of 36 not exceeding 91 is $36 \cdot 2 = 72$. Subtracting gives $r = 91 - 72 = 19$ (and $0 \leq 19 < 36$). Thus:
$$91 = 36 \cdot 2 + 19 \quad (q = 2, r = 19)$$
2. $\boldsymbol{a = 65, d = 12}$: Dividing 65 by 12, the largest multiple of 12 not exceeding 65 is $12 \cdot 5 = 60$. Subtracting gives $r = 65 - 60 = 5$ (and $0 \leq 5 < 12$). Thus:
$$65 = 12 \cdot 5 + 5 \quad (q = 5, r = 5)$$
3. $\boldsymbol{a = 121, d = 10}$: Dividing 121 by 10, the largest multiple of 10 not exceeding 121 is $10 \cdot 12 = 120$. Subtracting gives $r = 121 - 120 = 1$ (and $0 \leq 1 < 10$). Thus:
$$121 = 10 \cdot 12 + 1 \quad (q = 12, r = 1)$$

PITFALL / WARNING **Negative Dividends in the Division Algorithm**

In the Division Algorithm, the remainder r must **always** be non-negative: $0 \leq r < d$. For example, if $a = -11$ and $d = 3$, it is **incorrect** to write $-11 = 3(-3) - 2$ because $r = -2 < 0$ violates $0 \leq r < d$. Instead, we round the quotient **down** to $q = \left\lfloor -\frac{11}{3} \right\rfloor = -4$:

$$-11 = 3(-4) + 1 \implies q = -4, r = -11 \bmod 3 = 1$$

Note that in C, C++, and Java, `-11 % 3` evaluates to `-2` (truncated division toward zero), whereas Python's `-11 % 3` evaluates to `1` (true mathematical modulo). In discrete mathematics, r is always in the range $0 \leq r < d$.

2.1 The Modulo Operator and Congruence Relation

From the Division Algorithm $a = dq + r$ $(0 \leq r < d)$, the remainder r can be written using modulo notation:

$$r = a \bmod d \quad \text{or as a congruence} \quad r \equiv a \pmod{d}$$

EXAMPLE & WALKTHROUGH **Modulo Remainder Examples**

- $2 \equiv 5 \bmod 3$ because $5 = 3 \cdot 1 + 2$, so the remainder of 5 divided by 3 is $r = 2$.
- $1 \equiv 10 \bmod 3$ because $10 = 3 \cdot 3 + 1$, so the remainder of 10 divided by 3 is $r = 1$.
- $15 \equiv 35 \bmod 20$ because $35 = 20 \cdot 1 + 15$, so the remainder of 35 divided by 20 is $r = 15$.

3 Modular Arithmetic

Often in computer science and mathematics, we care only about the remainder left when an integer is divided by a fixed positive integer m, called the **modulus**. For example, a 12-hour or 24-hour clock wraps around modulo 12 or 24, and 32-bit unsigned integer registers wrap around modulo 2^{32}.

DEFINITION **Congruence Modulo m**

Let a and b be integers, and let m be a positive integer ($m \in \mathbb{Z}^+$). We say that $\boldsymbol{a}$ **is congruent to** $\boldsymbol{b}$ **modulo** $\boldsymbol{m}$, written:

$$a \equiv b \pmod{m}$$

if and only if m divides their difference, i.e., $m \mid (a - b)$ (or equivalently, $a = b + km$ for some $k \in \mathbb{Z}$). If $m \nmid (a - b)$, we write $a \not\equiv b \pmod{m}$.

CORE CONCEPT **Theorem: Congruence and Equal Remainders**

Let a and b be integers, and let m be a positive integer. Then:

$$a \equiv b \pmod{m} \iff a \bmod m = b \bmod m$$

That is, two integers are congruent modulo m if and only if they leave the **exact same remainder** when divided by m.

EXAMPLE & WALKTHROUGH **Congruence Examples**

- $5 \equiv 15 \pmod{10}$ because $5 \bmod 10 = 5$ and $15 \bmod 10 = 5$ (and $10 \mid (5 - 15) = -10$).
- $6 \equiv 21 \pmod{15}$ because $6 \bmod 15 = 6$ and $21 \bmod 15 = 6$ (and $15 \mid (6 - 21) = -15$).

INTUITION & DEEP DIVE **Operator vs. Relation: a mod m vs. a ≡ b (mod m)**

It helps to distinguish between the binary **operator** $a \bmod m$ and the equivalence **relation** $a \equiv b \pmod{m}$:

- **$a \bmod m$ is a function** from $\mathbb{Z} \times \mathbb{Z}^+$ to $\{0, 1, \ldots, m-1\}$ that returns a **single remainder** in $[0, m-1]$. For example, $15 \bmod 10 = 5$.
- **$a \equiv b \pmod m$ is a relation** on $\mathbb{Z}$ stating that a and b belong to the same remainder class. Thus $5 \equiv 15 \equiv 25 \equiv -5 \pmod{10}$ are all true at the same time.

3.1 Arithmetic Operations Modulo m

A very useful property of congruence modulo m is that it is compatible with addition, subtraction, and multiplication. This allows us to reduce intermediate results modulo m **before** adding or multiplying them, keeping numbers small and avoiding overflow.

CORE CONCEPT **Theorem & Corollary: Modular Addition and Multiplication**

Theorem: Let m be a positive integer. If $a \equiv b \pmod m$ and $c \equiv d \pmod m$, then:

$$a + c \equiv b + d \pmod m \quad \text{and} \quad ac \equiv bd \pmod m$$

Proof: Since $a \equiv b \pmod m$ and $c \equiv d \pmod m$, there exist integers $s, t \in \mathbb{Z}$ such that $a = b + sm$ and $c = d + tm$. Then:

- $a + c = (b + d) + (s + t)m \implies a + c \equiv b + d \pmod m$.
- $ac = (b + sm)(d + tm) = bd + m(bt + sd + stm) \implies ac \equiv bd \pmod m$. $\square$

Corollary: Let m be a positive integer and let a and b be integers. Then:

$$(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$$

$$ab \bmod m = ((a \bmod m)(b \bmod m)) \bmod m$$

EXAMPLE & WALKTHROUGH **Applying Modular Arithmetic**

1. **Combining Congruences**: Given $2 \equiv 5 \pmod 3$ and $1 \equiv 4 \pmod 3$:
   - Adding both sides gives $2 + 1 \equiv 5 + 4 \pmod 3$, i.e., $\mathbf{3 \equiv 9 \pmod 3}$ (both leave remainder 0).
   - Multiplying both sides gives $2 \cdot 1 \equiv 5 \cdot 4 \pmod 3$, i.e., $\mathbf{2 \equiv 20 \pmod 3}$ (both leave remainder 2, since $20 = 3 \cdot 6 + 2$).
2. **Modular Sum Reduction**: Compute $123 \bmod 11$ by splitting $123 = 100 + 23$:

$$\begin{aligned} 123 \bmod 11 &= ((100 \bmod 11) + (23 \bmod 11)) \bmod 11 \\ &= (1 + 1) \bmod 11 = 2 \end{aligned}$$

   (Check: $123 = 11 \cdot 11 + 2$.)
3. **Modular Product Reduction**: Compute $450 \bmod 21$ by factoring $450 = 90 \cdot 5$:

$$\begin{aligned} 450 \bmod 21 &= ((90 \bmod 21)(5 \bmod 21)) \bmod 21 \\ &= (6 \cdot 5) \bmod 21 = 30 \bmod 21 = 9 \end{aligned}$$

   (Check: $450 = 21 \cdot 21 + 9$.)

PITFALL / WARNING **Modular Division Cannot Be Done Directly**

While addition, subtraction, and multiplication work cleanly modulo m, **dividing (canceling a common factor) does not generally work**:

$$ac \equiv bc \pmod m \quad \not\Rightarrow \quad a \equiv b \pmod m$$

For example, $2 \cdot 3 \equiv 4 \cdot 3 (\text{mod } 6)$ (since $6 \equiv 12 (\text{mod } 6)$), but dividing both sides by 3 gives $2 \equiv 4 (\text{mod } 6)$, which is **false**. You can only cancel a common factor c modulo m when $\gcd(c, m) = 1$.

4 Prime Numbers & Primality Testing

Every integer greater than 1 is classified as either prime or composite based on its positive divisors.

DEFINITION **Prime and Composite Numbers**

- An integer $p > 1$ is called **prime** if and only if its only positive factors are 1 and p itself:

$$p \text{ is prime} \iff (p > 1) \land \forall a \in \mathbb{Z}^+ (a \mid p \implies (a = 1 \lor a = p))$$

- An integer $n > 1$ that is not prime is called **composite**. Equivalently, $n > 1$ is composite if and only if there exist integers a, b such that $n = ab$ with $1 < a < n$ and $1 < b < n$.
- **Note**: The integer 1 is **neither prime nor composite**. The first several primes are $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, \ldots$

4.1 The Fundamental Theorem of Arithmetic

Prime numbers serve as the multiplicative building blocks for all positive integers.

CORE CONCEPT **Theorem: The Fundamental Theorem of Arithmetic**

Every integer greater than 1 can be written **uniquely** as a prime or as the product of two or more primes, where the prime factors are written in order of nondecreasing size:

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k} \quad \text{with } p_1 < p_2 < \ldots < p_k \text{ primes and } a_i \in \mathbb{Z}^+$$

EXAMPLE & WALKTHROUGH **Prime Factorizations of Integers**

Find the prime factorizations of 100, 641, 999, and 1024:

- $100 = 2 \cdot 2 \cdot 5 \cdot 5 = 2^2 \cdot 5^2$
- $641 = 641$ (a prime number)
- $999 = 27 \cdot 37 = 3 \cdot 3 \cdot 3 \cdot 37 = 3^3 \cdot 37$
- $1024 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^{10}$

4.2 Primality Testing via the $\sqrt{n}$ Bound

To check whether an integer n (such as 641 above) is prime, a naive approach would test all integers from 2 up to $n - 1$. However, the following theorem cuts the work from $O(n)$ down to $O(\sqrt{n})$: we only need to test **prime** divisors up to $\lfloor \sqrt{n} \rfloor$.

CORE CONCEPT **Theorem: Prime Divisor Bound for Composite Numbers**

Theorem: If n is a composite integer, then n has a prime divisor less than or equal to $\sqrt{n}$.

Proof: Since n is composite, we can write $n = ab$ where $1 < a < n$ and $1 < b < n$. Without loss of generality, assume $a \leq b$. Multiplying both sides of $a \leq b$ by a gives:

$$a^2 \leq ab = n \quad \implies \quad a \leq \sqrt{n}$$

Since $a > 1$, by the Fundamental Theorem of Arithmetic, a is either prime itself or has a prime factor $p \leq a$. By transitivity of divisibility ($p \mid a$ and $a \mid n$), $p \mid n$ and $p \leq a \leq \sqrt{n}$. $\square$

Contrapositive — Primality Test: If an integer $n > 1$ has **no prime factor** less than or equal to $\sqrt{n}$, then n is **prime**.

EXAMPLE & WALKTHROUGH **Primality Testing Walkthrough**

1. **Testing $n = 21$**: We have $\sqrt{21} \approx 4.6$. The primes less than or equal to 4.6 are 2 and 3. Checking divisibility: $2 \nmid 21$, but $\mathbf{3 \mid 21}$ (since $21 = 3 \cdot 7$). Thus $n = 21$ is composite and has a prime factor $3 \leq \sqrt{21}$.
2. **Testing $n = 19$**: We have $\sqrt{19} \approx 4.36$. The only primes less than or equal to 4.36 are 2 and 3.
   - Does $2 \mid 19$? No ($19 = 2 \cdot 9 + 1$).
   - Does $3 \mid 19$? No ($19 = 3 \cdot 6 + 1$).

   Because 19 has no prime factor $\leq \sqrt{19}$, **19 is prime**.
3. **Testing $n = 641$**: Since $\sqrt{641} \approx 25.31$, we test all primes $p \leq 25$: $p \in \{2, 3, 5, 7, 11, 13, 17, 19, 23\}$. None of these nine primes divide 641, confirming that **641 is prime**.

5 Greatest Common Divisor & The Euclidean Algorithm

DEFINITION **Greatest Common Divisor (GCD)**

Let a and b be integers, not both zero. The largest integer d such that $d \mid a$ and $d \mid b$ is called the **greatest common divisor** of a and b, denoted $\gcd(a, b)$.

- **Relatively Prime (Coprime) Integers**: Two integers a and b are called **relatively prime** if their greatest common divisor is 1, i.e., $\gcd(a, b) = 1$.

EXAMPLE & WALKTHROUGH **GCD and Relatively Prime Examples**

- $\mathbf{gcd(6, 9) = 3}$: The positive divisors of 6 are $\{1, 2, 3, 6\}$ and of 9 are $\{1, 3, 9\}$. Their common divisors are $\{1, 3\}$, and the largest is 3.
- $\mathbf{gcd(24, 36) = 12}$: The common positive divisors of 24 and 36 are $\{1, 2, 3, 4, 6, 12\}$, so the greatest is 12.
- **5 and 7 are relatively prime**: $\gcd(5, 7) = 1$.
- **12 and 25 are relatively prime**: Even though both $12 = 2^2 \cdot 3$ and $25 = 5^2$ are composite, they share no prime factors, so $\gcd(12, 25) = 1$.

INTUITION & DEEP DIVE **Finding GCD and LCM from Prime Factorizations**

When the prime factorizations of a and b are already known:

$$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}, \quad b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n} \quad (a_i, b_i \geq 0)$$

we can find both the Greatest Common Divisor (gcd) and the Least Common Multiple (lcm) by taking the minimum and maximum exponents of each prime:

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \cdots p_n^{\min(a_n, b_n)}, \quad \operatorname{lcm}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \cdots p_n^{\max(a_n, b_n)}$$

Since $\min(x, y) + \max(x, y) = x + y$, these satisfy the useful identity $\boldsymbol{ab = \gcd(a, b) \cdot \operatorname{lcm}(a, b)}$. However, factoring large integers is computationally expensive. Instead, the **Euclidean Algorithm** computes $\gcd(a, b)$ in $O(\log \min(a, b))$ division steps without factoring either number.

5.1 The Euclidean Algorithm

The Euclidean Algorithm is based on a simple observation: if $a = bq + r$, then any common divisor of a and b also divides $r = a - bq$, and vice versa. Therefore:

$$\gcd(a, b) = \gcd(b, a \bmod b)$$

Algorithm: The Euclidean Algorithm `O(log b) division steps`

Input: Positive integers a and b with $a > b > 0$.
Output: The greatest common divisor $\gcd(a, b)$.

1. Write a and b in the Division Algorithm form:
$$a = b \cdot q + r \quad \text{where } 0 \leq r < b$$
2. If $r = 0$, terminate and output $\gcd = b$.
3. Otherwise, replace $a \leftarrow b$ and $b \leftarrow r$, and repeat from Step 1 until r becomes 0.
4. The $\gcd(a, b)$ is the value of b when $r = 0$ (equivalently, the **last nonzero remainder**).

EXAMPLE & WALKTHROUGH **Euclidean Algorithm Trace: gcd(414, 662)**

Problem: Find the greatest common divisor of 414 and 662 using the Euclidean algorithm.

Solution: Setting $a = 662$ and $b = 414$, we repeatedly apply the Division Algorithm:

Step	Division Equation ($a = b \cdot q + r$)	a	b	Remainder r & Next Step
1	$662 = 414 \cdot 1 + 248$	662	414	$r = 248 \neq 0 \Longrightarrow (a, b) \leftarrow (414, 248)$
2	$414 = 248 \cdot 1 + 166$	414	248	$r = 166 \neq 0 \Longrightarrow (a, b) \leftarrow (248, 166)$
3	$248 = 166 \cdot 1 + 82$	248	166	$r = 82 \neq 0 \Longrightarrow (a, b) \leftarrow (166, 82)$
4	$166 = 82 \cdot 2 + 2$	166	82	$r = 2 \neq 0 \Longrightarrow (a, b) \leftarrow (82, 2)$
5	$82 = 2 \cdot 41 + 0$	82	2	$r = 0 \Longrightarrow$ Stop! Output $b = 2$

Hence, $\mathbf{\gcd(414, 662) = 2}$, because 2 is the last nonzero remainder.

6 Worked Practice Exercises

Below are complete worked solutions to the practice problems from Kenneth H. Rosen's **Discrete Mathematics and Its Applications** (Section 4.1 on Divisibility and Modular Arithmetic, and Section 4.3 on Primes and Greatest Common Divisors).

EXAMPLE & WALKTHROUGH **Divisibility and Modular Arithmetic (Section 4.1, Nos. 1, 9, 10, 11, 12, 20, 30)**

1. Does 17 divide each of these numbers?

- **a) 68**: Yes, $17 \mid 68$ because $68 = 17 \cdot 4$.
- **b) 84**: No, $17 \nmid 84$ because $84 = 17 \cdot 4 + 16$ ($r = 16 \neq 0$).
- **c) 357**: Yes, $17 \mid 357$ because $357 = 17 \cdot 21$.
- **d) 1001**: No, $17 \nmid 1001$ because $1001 = 17 \cdot 58 + 15$ ($r = 15 \neq 0$).

9–12. Find the quotient q and remainder r when:

No.	Problem (a divided by d)	Division Algorithm ($a = dq + r, 0 \leq r < d$)	Quotient q	Remainder r
9a	19 divided by 7	$19 = 7 \cdot 2 + 5$	2	5
9b	-111 divided by 11	$-111 = 11 \cdot (-11) + 10$	-11	10
9c	789 divided by 23	$789 = 23 \cdot 34 + 7$	34	7
9d	1001 divided by 13	$1001 = 13 \cdot 77 + 0$	77	0
9e	0 divided by 19	$0 = 19 \cdot 0 + 0$	0	0
9f	3 divided by 5	$3 = 5 \cdot 0 + 3$	0	3
9g	-1 divided by 3	$-1 = 3 \cdot (-1) + 2$	-1	2
9h	4 divided by 1	$4 = 1 \cdot 4 + 0$	4	0
10a	44 divided by 8	$44 = 8 \cdot 5 + 4$	5	4
10b	777 divided by 21	$777 = 21 \cdot 37 + 0$	37	0
10c	-123 divided by 19	$-123 = 19 \cdot (-7) + 10$	-7	10
10d	-1 divided by 23	$-1 = 23 \cdot (-1) + 22$	-1	22
10e	-2002 divided by 87	$-2002 = 87 \cdot (-24) + 86$	-24	86
10f	0 divided by 17	$0 = 17 \cdot 0 + 0$	0	0
10g	$1,234,567$ divided by 1001	$1,234,567 = 1001 \cdot 1233 + 334$	1233	334
10h	-100 divided by 101	$-100 = 101 \cdot (-1) + 1$	-1	1

- **11. Find $-101 \bmod 13$ and $199 \bmod 19$**: Since $-101 = 13 \cdot (-8) + 3$, we have $-101 \bmod 13 = 3$. Since $199 = 19 \cdot 10 + 9$, we have $199 \bmod 19 = 9$.
- **12. Find $-88 \bmod 13$ and $288 \bmod 17$**: Since $-88 = 13 \cdot (-7) + 3$, we have $-88 \bmod 13 = 3$. Since $288 = 17 \cdot 16 + 16$, we have $288 \bmod 17 = 16$.

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20. Determine whether each of these integers is congruent to 5 modulo 17:

- **a) 80**: $80 - 5 = 75$, and $17 \nmid 75$ ($75 = 17 \cdot 4 + 7$). Thus $80 \not\equiv 5 \pmod{17}$ (in fact, $80 \bmod 17 = 12$).
- **b) 103**: $103 - 5 = 98$, and $17 \nmid 98$ ($98 = 17 \cdot 5 + 13$). Thus $103 \not\equiv 5 \pmod{17}$ (in fact, $103 \bmod 17 = 1$).
- **c) -29**: $-29 - 5 = -34 = 17 \cdot (-2)$, so $17 \mid (-29 - 5)$. Thus $\mathbf{-29 \equiv 5 \pmod{17}}$.
- **d) -122**: $-122 - 5 = -127$, and $17 \nmid -127$ ($-127 = 17(-8) + 9$). Thus $-122 \not\equiv 5 \pmod{17}$.

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30. Evaluate each of these quantities using modular arithmetic properties:

- **a) $(-133 \bmod 23 + 261 \bmod 23) \bmod 23$**: This simplifies to $(-133 + 261) \bmod 23 = 128 \bmod 23$. Since $128 = 23 \cdot 5 + 13$, the result is $\mathbf{13}$.
- **b) $(457 \bmod 23 \cdot 182 \bmod 23) \bmod 23$**: First reduce each factor: $457 = 23 \cdot 19 + 20 \implies 457 \bmod 23 = 20 \equiv -3 \pmod{23}$, and $182 = 23 \cdot 7 + 21 \implies 182 \bmod 23 = 21 \equiv -2 \pmod{23}$. Multiplying gives $(20 \cdot 21) \bmod 23 = ((-3)(-2)) \bmod 23 = 6 \bmod 23 = \mathbf{6}$.
- **c) $(99^2 \bmod 32)^3 \bmod 15$**: First compute $99 \bmod 32 = 3$ (since $99 = 32 \cdot 3 + 3$). Thus $99^2 \bmod 32 = 3^2 \bmod 32 = 9$. Next, evaluate $9^3 \bmod 15$: since $9^2 = 81 = 15 \cdot 5 + 6 \implies 9^2 \bmod 15 = 6$, we have $9^3 \bmod 15 = (6 \cdot 9) \bmod 15 = 54 \bmod 15 = \mathbf{9}$.

- **d) $((34 \cdot 102) \bmod 27)^2 \bmod 13$**: Reduce modulo 27: $34 \bmod 27 = 7$ and $102 = 27 \cdot 3 + 21 \implies 102 \bmod 27 = 21 \equiv -6 (\bmod 27)$. Thus $(34 \cdot 102) \bmod 27 = (7 \cdot (-6)) \bmod 27 = -42 \bmod 27 = 12$. Finally, $12^2 \bmod 13 = (-1)^2 \bmod 13 = \mathbf{1}$.

EXAMPLE & WALKTHROUGH **Primes and Greatest Common Divisors (Section 4.3, Nos. 1, 14, 15, 32)**

1. Determine whether each of these integers is prime:

- **a) 21**: Composite, since $3 \mid 21$ ($21 = 3 \cdot 7$).
- **b) 29**: We test primes $p \leq \sqrt{29} \approx 5.38$, i.e., $p \in \{2, 3, 5\}$. Since $2 \nmid 29$, $3 \nmid 29$, and $5 \nmid 29$, **29 is prime**.
- **c) 71**: We test primes $p \leq \sqrt{71} \approx 8.42$, i.e., $p \in \{2, 3, 5, 7\}$. None divide 71, so **71 is prime**.
- **d) 97**: We test primes $p \leq \sqrt{97} \approx 9.85$, i.e., $p \in \{2, 3, 5, 7\}$. None divide 97, so **97 is prime**.
- **e) 111**: Composite. The sum of its digits is $1 + 1 + 1 = 3$, so $3 \mid 111$ ($111 = 3 \cdot 37$).
- **f) 143**: We test primes $p \leq \sqrt{143} \approx 11.95$, i.e., $p \in \{2, 3, 5, 7, 11\}$. Checking 11: $143 = 11 \cdot 13$, so **143 is composite**.

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14. Which positive integers less than 12 are relatively prime to 12? Since $12 = 2^2 \cdot 3$, an integer k satisfies $\gcd(k, 12) = 1$ if and only if $2 \nmid k$ and $3 \nmid k$. Checking $k \in \{1, 2, \ldots, 11\}$:

- Multiples of 2: 2, 4, 6, 8, 10 (excluded)
- Multiples of 3: 3, 6, 9 (excluded)

The remaining integers relatively prime to 12 are $\mathbf{1, 5, 7, 11}$.

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15. Determine whether the following sets of integers are pairwise relatively prime:

- **a) 21, 34, 55**: Prime factorizations are $21 = 3 \cdot 7$, $34 = 2 \cdot 17$, and $55 = 5 \cdot 11$. Since no two numbers share a prime factor, $\gcd(21, 34) = \gcd(21, 55) = \gcd(34, 55) = 1$. **Yes, pairwise relatively prime.**
- **b) 14, 17, 85**: Although $\gcd(14, 17) = 1$ and $\gcd(14, 85) = 1$, we have $\gcd(17, 85) = 17 \neq 1$ (since $85 = 5 \cdot 17$). **No, not pairwise relatively prime.**
- **c) 25, 41, 49, 64**: Prime factorizations are $25 = 5^2$, $41 = 41$, $49 = 7^2$, and $64 = 2^6$. All prime bases $(5, 41, 7, 2)$ are distinct. **Yes, pairwise relatively prime.**
- **d) 17, 18, 19, 23**: Since 17, 19, 23 are distinct primes that do not divide $18 = 2 \cdot 3^2$, every pair has $\gcd = 1$. **Yes, pairwise relatively prime.**

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32. Use the Euclidean algorithm to find the greatest common divisor of each pair:

- **a) $\gcd(1, 5)$**: $5 = 1 \cdot 5 + 0 \implies \gcd(1, 5) = 1$.
- **b) $\gcd(100, 101)$**: $101 = 100 \cdot 1 + 1$, then $100 = 1 \cdot 100 + 0 \implies \gcd(100, 101) = 1$.
- **c) $\gcd(123, 277)$**:
  - $277 = 123 \cdot 2 + 31$
  - $123 = 31 \cdot 3 + 30$
  - $31 = 30 \cdot 1 + 1$
  - $30 = 1 \cdot 30 + 0$

  Hence $\gcd(123, 277) = 1$.
- **d) $\gcd(1529, 14039)$**:
  - $14039 = 1529 \cdot 9 + 278$
  - $1529 = 278 \cdot 5 + 139$
  - $278 = 139 \cdot 2 + 0$

  Hence $\gcd(1529, 14039) = 139$.
- **e) $\gcd(1529, 14038)$**:

- $14038 = 1529 \cdot 9 + 277$
- $1529 = 277 \cdot 5 + 144$
- $277 = 144 \cdot 1 + 133$
- $144 = 133 \cdot 1 + 11$
- $133 = 11 \cdot 12 + 1$
- $11 = 1 \cdot 11 + 0$

Hence $\mathbf{gcd(1529, 14038) = 1}$.

- **f)** $\mathbf{gcd(11111, 111111)}$:
  - $111111 = 11111 \cdot 10 + 1$
  - $11111 = 1 \cdot 11111 + 0$

Hence $\mathbf{gcd(11111, 111111) = 1}$.